Is Statistical Process Control Obsolete For Quality Control In ELECTRONICS MANUFACTURING?

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Quality influencing actions come from informed decisions. You cannot fix what you do not measure

Statistical process control (SPC) has long been an important tactic for companies looking to ensure high product quality. In modern electronics manufacturing, complexities involved do not comply with the fundamental assumption of process stability. This makes traditional SPC worthless as a high-level approach to quality management, when combined with the increasing amount of data collected. An approach compliant with Lean Six Sigma philosophy, allowing a wider scope than SPC, is superior in identifying and prioritising relevant improvement initiatives.

SPC was introduced in the 1920s, designed to address manufacturing of that era. One purpose of SPC was early detection of undesired behaviour, allowing for early intervention and improvement. Limitations were set by then available IT, a landscape completely different from modern times.

Back-tracking Moore's Law, it is easy to accept that not only IT, but also product complexities and capabilities were different than today. In fact, measurements from manufacturing operations have no common measure with today's situation. Following this complexity, combined with factors such as globalised markets driving up volume of manufacturing, the result is that the amount of output data today is incomprehensible by 1920 standards.

Fundamental limitations

SPC appears to still hold an important position for original equipment manufacturers (OEMs). It is found in continuous manufacturing processes, calculating control limits and attempting to detect out-of-order process parameters. In theory, such control limits help visualise if things are turning from good to worse. But theory is, well, theory!

A fundamental assumption of SPC is that you can remove or account for common cause variations from the process, which means all variations remaining are special cause ones. These are the parameters you need to worry about when these start to drift.

An electronics product today contains hundreds of components. It experiences many design modifications due to things such as component obsolescence. It is tested in various stages during assembly process, features multiple firmware revisions, test software versions, test operators, variance in environmental factors and so on.

Example of high dynamics

An example of high dynamics is Aidon, manufacturer of smart metering products. According to Aidon's head of production, Petri Ounila, an average production batch contains 10,000 units, has units containing over 350 electronics components each and experiences more than 35 component changes throughout this build process.

This gives Aidon a new product, or process, every 280th unit. In addition, there are changes to test processes, fixtures, test programs, instrumentation and more. The result is an estimated average of a new process every 10th unit, or less. Or in other words, there are 1000 different processes in manufacturing a single batch.

How would you even begin to eliminate common cause variations here? And should you even do so?

Even if you managed, how would you go about implementing the alarming system?

A method in SPC, developed by Western Electric Co. back in 1956, is known as Western Electrical Rules, or WECO. It specifies

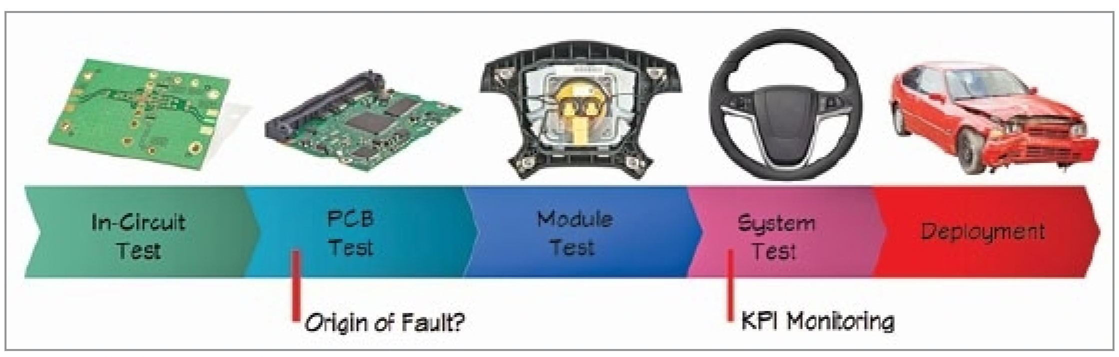

Fig. 1: Explaining KPIs